

Global Markets Intelligence Dashboard

A Multi-Market Financial Analytics & Signal Confirmation System

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Document Purpose

This document is the single source of truth for the Global Markets Intelligence Dashboard project. It defines scope, data sources, technical architecture, the signal confirmation system, and stated limitations. All technical claims in this document (library names, API free-tier terms, ticker symbols) were verified against current documentation at the time of writing, not assumed from memory. Where a term of service or usage restriction applies, it is stated explicitly rather than glossed over.

1. Project Overview

Retail investors and analysts in Pakistan lack a single, accessible tool to compare risk-adjusted performance and signal-based trade opportunities across local equities, cryptocurrency, precious metals, and global equities. This project builds that tool using exclusively free, publicly accessible data sources, deployed as a live, interactive web dashboard.

The project is intentionally scoped as a **portfolio and learning artifact**, not a production trading system. It is built by a second-year BS Data Science student following completion of the IBM Data Analyst Professional Certificate. Its goal is to demonstrate applied data analysis, financial reasoning, software architecture, and honest self-assessment of a junior-level project's limitations — the combination of which is what differentiates it from template-based portfolio projects.

1.1 What This Project Is

- A multi-market financial data pipeline covering four asset classes.
- A reusable analytical framework (returns, volatility, correlation, risk-adjusted performance).
- A two-layer signal confirmation system combining technical indicators with candlestick pattern recognition.
- A historical backtesting module for all four markets.
- A paper-trading extension (simulated capital, real-time data) for the two markets where free paper-trading infrastructure exists.
- A live, publicly deployed, multi-page interactive dashboard.

1.2 What This Project Is Not

- Not a live automated trading system connected to real brokerage capital.
- Not a statistically validated trading strategy — sample sizes at this stage are small by design and stated as such.
- Not a claim of quantitative trading expertise — it is a demonstrated foundation, one deliverable within a longer-term learning path toward that goal.

2. Scope and Execution Phases

The project is executed in four sequential phases. Order is deliberate: Phase 1 (PSX) is the least standardized data source and the most differentiating market, so the reusable architecture is built against the hardest case first. Later phases reuse that architecture, adding only the new data integration and market-specific adjustments each requires.

Phase	Market	Primary Data Source	Rationale for Position
1	Pakistan Stock Exchange (PSX)	psxdata (open-source Python library)	Hardest integration; the project's key differentiator; architecture built here first.
2	Cryptocurrency	CoinGecko API (Demo/free tier)	Cleanest, best-documented API; fastest to integrate once architecture exists.
3	Precious Metals (Gold, Silver)	yfinance (GC=F, SI=F futures tickers)	Shares a data pipeline pattern with Phase 4.

Phase	Market	Primary Data Source	Rationale for Position
4	Global Equities	yfinance	Same library as Phase 3; minimal new integration work.

3. Core Analytical Framework (Shared Across All Phases)

Every market phase computes the same four base metrics. This is the reusable core of the project — built once in a shared module, applied identically regardless of asset class.

- **Returns** — period-over-period percentage change in price.
- **Volatility** — standard deviation of returns over a rolling window.
- **Correlation** — pairwise correlation between assets within a given market.
- **Risk-adjusted performance** — a simplified Sharpe-style ratio (mean return divided by volatility), used to compare assets on a like-for-like basis rather than by raw return alone.

4. Two-Layer Signal Confirmation System

Rather than generating trade signals from a single indicator (a common and weaker approach), the system requires two independent layers to agree before a signal is considered confirmed. This is a recognized technique in technical analysis known as signal confluence, and it reduces false-positive signals at the cost of generating fewer total signals overall — a tradeoff stated explicitly in Section 6.

4.1 Layer 1 — Indicator Condition (the Setup)

Computed via technical indicators: Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Bollinger Bands. Example: RSI falling below 30 flags a potential oversold condition — a candidate setup, not yet a signal.

4.2 Layer 2 — Candlestick Pattern Confirmation (the Trigger)

A recognized candlestick reversal or continuation pattern (e.g. Hammer, Bullish Engulfing, Doji) must appear within a defined confirmation window following the Layer 1 condition. Only when both layers align does the system emit a confirmed BUY, SELL, or HOLD signal.

4.3 Confirmation Window

The number of bars within which a Layer 2 pattern must appear after a Layer 1 condition is a tunable parameter, not a fixed default. It will be tested and justified with reasoning in the project write-up rather than left at an arbitrary value.

4.4 Implementation Library

Indicator and candlestick pattern computation uses **pandas-ta-classic**, a pure-Python library offering 224+ indicators and 62 native candlestick patterns with no dependency on TA-Lib's C library. This choice is deliberate: TA-Lib requires a compiled C dependency that is a documented source of installation friction on Windows; pandas-ta-classic avoids this entirely while covering the same pattern set.

5. Backtesting and Paper Trading Scope

Every phase includes historical backtesting of the confirmed-signal strategy. Paper trading (simulated capital, real-time market data, full order-execution logic) is added where free infrastructure genuinely supports it.

Phase / Market	Historical Backtest	Paper Trading	Notes
PSX	Yes	Not available	No free paper-trading broker covers PSX-listed instruments.
Cryptocurrency	Yes	Yes — via Alpaca	Alpaca paper accounts are free and open to signup globally.
Precious Metals	Yes	Not available	Alpaca does not support commodity/futures paper trading.
Global Equities	Yes	Yes — via Alpaca	Same Alpaca paper account covers US-listed equities.

Paper trading via Alpaca uses simulated capital against live, real-time market data and full order execution logic (market, limit, stop orders). No real capital is placed at risk at any point in this project. The architecture separates signal generation from execution logic, meaning the same signal code could in principle point at a live account instead of a paper one — this is a statement about clean software design, not an intention to trade with real capital as part of this project.

6. Stated Limitations and Disclosures

The following limitations are disclosed deliberately and in advance, rather than surfacing only if questioned. This section is a required part of every phase's final write-up.

Limitation 1 — Sample Size

Confirmed two-layer signals are, by design, less frequent than single-layer signals. Trade counts per market over the test period will be small. Win rates calculated on a small sample are directional evidence, not statistical proof of an edge, and will be labeled as such in every results section.

Limitation 2 — No Cross-Market Pooling of Results

PSX, cryptocurrency, precious metals, and global equities are not statistically comparable populations — they differ in volatility regime, trading hours, and liquidity. Results are reported separately per market. A combined figure, if shown at all, is clearly labeled as an approximate summary, not a headline finding.

Limitation 3 — No Out-of-Sample Validation at This Stage

Signal thresholds and confirmation windows are being designed against the same historical data used to evaluate them in this phase of the project. This is disclosed as a known gap. A more rigorous version of this project would split each market's data into a design period and a separate, untouched test period.

Limitation 4 — PSX Data Usage Terms

The Pakistan Stock Exchange's public website terms restrict commercial redistribution of its market data feed. This project uses publicly available data for personal, educational, and portfolio purposes only, with no redistribution or commercial resale of the data itself. A disclaimer to this effect is included in the project's public README.

7. Technology Stack

Layer	Tool / Library	Purpose
Data — PSX	psxdata	Free, open-source, no API key; historical + live PSX data as pandas DataFrames.
Data — Crypto	CoinGecko API (Demo tier)	Free registration, no card; 10,000 calls/month, 100 calls/min, 1 year of daily/hourly historical data.
Data — Metals & Global Stocks	yfinance	Free; gold (GC=F) and silver (SI=F) futures plus global equity tickers.
Analysis	pandas, numpy	Data wrangling, returns/volatility/correlation calculations.
Indicators & Patterns	pandas-ta-classic	224+ indicators, 62 native candlestick patterns, no TA-Lib/C dependency.
Backtesting	Custom Python logic	Signal evaluation against historical data per market.
Paper Trading	Alpaca Trading API (paper mode)	Free, globally accessible paper accounts; real-time data; simulated capital only.
Visualization	Plotly	Interactive charts embedded in the dashboard.
Dashboard	Streamlit (multi-page app)	Python-native UI; no separate frontend framework required.
Hosting	Streamlit Community Cloud	Free hosting, auto-deploys from connected GitHub repository.
Version Control	GitHub	Single repository; one commit per work session, notebooks downloaded and committed immediately after each session.

8. Deliverable Structure

Repository layout:

global-markets-dashboard/

■■■ app.py (home / landing page)

■■■ pages/

■■■ 1_PSX.py

■■■ 2_Crypto.py

■■■ 3_Metals.py

■■■ 4_Global_Stocks.py

■■■ utils/

■■■ analysis.py (shared returns / volatility / correlation / indicator / signal functions)

■■■ data/

■■■ requirements.txt

■■■ README.md (includes limitations section and PSX data-use disclaimer)

9. Success Criteria

A phase is considered complete only when it produces one specific, defensible, plain-English insight grounded in the actual computed data for that market — not a generic statement that could apply to any dataset.

Example of an acceptable finding: "Sector X delivered the best risk-adjusted return over the test period despite Sector Y showing higher raw returns, because Y's volatility was substantially higher." This ties a conclusion to specific, checkable numbers from the project's own output.

10. Timeline

Timeline reflects current progress as of this document's writing: the IBM Data Analyst Professional Certificate (all 11 courses) was completed well ahead of its original schedule. The remaining roadmap allocates 1-2 hours per day to the Microsoft Power BI certificate alongside dedicated project time.

Phase	Estimated Duration	Notes
Phase 1 — PSX	6 days	Includes data pipeline, core framework, signal system, backtest, dashboard page.
Phase 2 — Crypto	2-3 days	Shorter — reuses Phase 1 architecture; adds Alpaca paper trading.
Phase 3 — Metals	1-2 days	Shares yfinance pipeline pattern with Phase 4.
Phase 4 — Global Equities	1-2 days	Adds Alpaca paper trading for this market.
Ongoing, parallel	1-2 hrs/day	Microsoft Power BI Professional Certificate, run alongside all project phases.

This timeline will be reviewed and explicitly adjusted, not silently extended, if any phase runs materially over its estimate.

11. Approval

This charter reflects the scope and design agreed upon prior to Phase 1 execution. Any material change in scope, data source, or timeline during execution should be reflected in a revised version of this document, not left undocumented.

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